

The Impact of Broad Money Growth on Inflation

For MENA Countries

2010–2024

Research focus

This study examines the relationship between broad money growth and inflation in seven selected MENA economies using panel-data econometrics. The analysis integrates macroeconomic theory, descriptive statistics, robust econometric inference, exchange-rate dynamics, sensitivity analysis, and Power BI data visualization.

 Countries	Algeria, Egypt, Jordan, Morocco, Saudi Arabia, Tunisia, Türkiye
 Period	2010–2024
 Dataset	105 raw observations; 98 observations in the regression sample
 Method	Panel Data Econometrics, Pooled OLS, Fixed Effects, Random Effects, Robust Inference
 Visualization	Power BI + R

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Abstract

This report evaluates the relationship between broad money growth and inflation in seven selected MENA economies—Algeria, Egypt, Jordan, Morocco, Saudi Arabia, Tunisia, and Türkiye—over 2010–2024. The empirical workflow begins with 105 raw country-year observations and uses 98 observations from 2011–2024 because the constructed depreciation rate requires the previous year’s exchange rate.

The reported model is pooled OLS, selected by the formal panel-effect tests, with heteroskedasticity-, serial-correlation-, and cross-sectional-dependence-robust inference. The central pooled estimate indicates that a one-percentage-point increase in broad money growth is associated with approximately a 0.436 percentage-point increase in inflation, holding GDP growth and currency depreciation constant.

Currency depreciation is also positively and significantly associated with inflation, while GDP growth is statistically insignificant.

However, the pooled monetary result is strongly sensitive to Türkiye: when Türkiye is excluded, the broad-money coefficient falls from 0.436 to 0.055 and its confidence interval includes zero. The evidence therefore supports a positive monetary association at the pooled regional level, but it does not establish a stable causal elasticity applicable uniformly across the MENA sample.

The report recommends treating money growth as one part of a broader inflation process involving exchange-rate pass-through, expectations, fiscal conditions, supply shocks, and country-specific monetary institutions.

Keywords: inflation, broad money, M2, MENA, panel data, pooled OLS, exchange-rate pass-through, monetary policy, robust standard errors.

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1. Executive Summary & Problem Definition

1.1. The core economic problem

The problem evaluate a central macroeconomic question: *to what extent is inflation in selected MENA economies associated with the growth of broad money, after controlling for real economic activity and currency depreciation?*

The question is economically important because broad money contains deposits and other liquid monetary instruments that finance transactions, working capital, household spending, and asset purchases. Rapid expansion of broad money can therefore increase nominal demand, especially when monetary expansion is not matched by a proportional increase in real output.

The empirical evidence in the dashboard suggests a strong positive pooled relationship. The average inflation rate over the regression sample is 7.69%, while average broad-money growth is 11.88%. Across countries, Türkiye records by far the highest mean inflation (21.88%) and the highest mean broad-money growth (29.13%). Egypt is second in inflation (14.30%) and also exhibits high average money growth (19.64%). At the opposite end, Morocco’s mean inflation is only 1.89%, with mean broad-money growth of 5.53%.

These cross-country contrasts already indicate that monetary conditions matter, but they also show that institutional and structural differences are large.

The main pooled regression estimated in the supplied R workflow is

$$\pi_{it} = \alpha + \beta_1 M_{it} + \beta_2 g_{it} + \beta_3 d_{it} + \varepsilon_{it}, \quad (1)$$

where π_{it} is CPI inflation, M_{it} is broad-money growth, g_{it} is real GDP growth, and d_{it} is the annual currency depreciation rate against the US dollar.

Using country-clustered HC3 standard errors, the estimated coefficients are:

Table 1: Main pooled OLS estimates with HC3 country-clustered standard errors

Variable	Coefficient	Std. Error	t	p -value	95% CI
Broad money growth	0.436	0.062	7.04	< 0.001	[0.557 , 0.315]
GDP growth	-0.148	0.240	-0.61	0.541	[0.324 , -0.619]
Depreciation rate	0.377	0.086	4.36	< 0.001	[0.546 , 0.207]

The same point estimates remain statistically significant for broad money and depreciation under Driscoll–Kraay standard errors.

The money coefficient is 0.436 with a standard error of 0.100 and $p < 0.001$; the depreciation

coefficient is 0.377 with a standard error of 0.074 and $p < 0.001$. GDP growth remains insignificant.

1.2. Key stakeholders

The relevant stakeholders are households, firms, commercial banks, central banks, finance ministries, exchange-rate authorities, and external investors.

Households care about purchasing power and inflation expectations. Firms face changes in input costs, financing conditions, and demand. Banks transmit monetary and credit conditions into the economy. Central banks are responsible for price stability and liquidity conditions. Fiscal authorities influence aggregate demand, deficits, public debt, and in some cases the incentives for monetary financing.

External investors react to exchange-rate risk, inflation expectations, and the credibility of macroeconomic policy.

1.3. Market failures and institutional frictions

Inflation is not best understood as a conventional microeconomic market failure such as a single monopoly or a standard information problem. The main relevant frictions are macroeconomic.

Nominal rigidities can prevent prices and wages from adjusting instantly. Monetary and fiscal institutions may face credibility problems, creating unanchored inflation expectations. Exchange-rate movements can generate imported inflation through energy, food, intermediate goods, and imported capital goods.

Credit-market frictions can make the response of spending to liquidity expansion asymmetric across sectors and households. In some countries, administered prices, subsidies, regulated exchange rates, and shallow financial markets can further weaken a simple textbook monetary transmission mechanism.

1.4. Scope and central conclusion

The analysis covers seven countries and 14 usable years per country in the regression sample. It is a small- N , short- T panel.

This matters: the dataset is sufficient for a useful comparative empirical study, but it is not large enough to justify strong causal claims.

The most important empirical conclusion is therefore two-sided:

1. At the pooled regional level, higher broad-money growth is strongly positively associated with inflation.
2. The association is not stable when Türkiye is removed, so the regional estimate should not be interpreted as a homogeneous MENA causal effect.

The second point is essential for an economically credible interpretation of the dashboard.

2. Logical Framework & Qualitative Analysis

2.1. Quantity Theory of Money

The theoretical starting point is the quantity equation

$$MV = PY, \quad (2)$$

where M is the money stock, V is velocity, P is the price level, and Y is real output.

Taking logarithms and first differences gives the approximate growth relationship

$$\mu_t + v_t \approx \pi_t + g_t, \quad (3)$$

so that

$$\pi_t \approx \mu_t + v_t - g_t. \quad (4)$$

Here μ_t is money growth, v_t is velocity growth, π_t is inflation, and g_t is real GDP growth.

Equation (4) provides the economic intuition for the positive money-growth coefficient.

If broad money grows faster while velocity and real output are unchanged, nominal expenditure must grow faster, creating upward pressure on prices.

However, the equation also shows why the regression coefficient need not equal one.

Velocity can change, financial intermediation can change the transaction content of money, and supply constraints can change the mapping from nominal expenditure into prices.

2.2. Aggregate demand, nominal rigidities, and the short run

In the short run, an increase in liquidity can stimulate consumption, investment, and credit before all prices adjust.

If firms operate with spare capacity, part of the response may initially appear as higher real output. As capacity constraints become binding, more of the nominal stimulus can be translated into prices.

Consequently, the money-inflation relationship can be nonlinear and state-dependent.

This mechanism is consistent with the project's GDP coefficient being negative but statistically insignificant.

The sign is compatible with a simple demand-capacity story in which stronger real output absorbs some nominal expansion, but the data do not provide sufficient evidence for a robust conditional GDP effect in this small panel.

2.3. Exchange-rate pass-through

The depreciation variable is theoretically important because the MENA economies are open to imported goods, energy, food, capital equipment, and intermediate inputs.

A depreciation raises domestic-currency import prices, which can propagate through producer costs and consumer prices.

The empirical estimate of 0.377 is positive and highly significant.

A simple pass-through relationship can be written as

$$\Delta p_t = \lambda d_t + \text{other cost pressures}, \quad \lambda > 0, \quad (5)$$

where d_t is the percentage depreciation of the domestic currency.

The positive regression estimate is consistent with this channel.

The dashboard also highlights a major institutional distinction.

Jordan and Saudi Arabia have essentially zero measured depreciation throughout the sample because their exchange rates are pegged.

Türkiye, by contrast, exhibits extremely large depreciation episodes; its average depreciation is 26.00%, versus 1.37% in Morocco and 0% in Jordan and Saudi Arabia.

2.4. Expectations, credibility, and policy regimes

The same amount of monetary expansion can have different inflation effects depending on expectations.

If households and firms believe the central bank will reverse excessive liquidity growth, velocity may adjust without a large persistent inflation response.

If policy credibility is weak, the same monetary expansion can be accompanied by currency substitution, faster turnover of money, wage-price adjustments, and a larger inflation response.

This helps explain why a common coefficient across seven countries is restrictive.

The countries in the sample do not share the same exchange-rate regime, fiscal institutions, financial depth, or inflation history.

2.5. Short-run versus long-run effects

The short-run effect of money growth can be partly absorbed by output, velocity, or portfolio reallocation.

Over a sufficiently long horizon, however, persistent nominal monetary expansion that is not matched by real output growth is more difficult to absorb without a corresponding increase in the price level.

The dataset is annual and covers only fourteen usable years per country, so it is better suited to identifying medium-run conditional associations than to tracing a full long-run monetary equilibrium.

3. Mathematical & Quantitative Formulation

3.1. Equilibrium framework

A compact macroeconomic representation is obtained by combining the quantity equation with an exchange-rate channel.

Let nominal expenditure be $E_t = P_t Y_t$ and let broad money growth be μ_t .

Then

$$\Delta \ln P_t = \Delta \ln M_t + \Delta \ln V_t - \Delta \ln Y_t. \quad (6)$$

If velocity growth depends on financial conditions, expectations, and exchange-rate movements,

$$\Delta \ln V_t = \phi_0 + \phi_1 d_t + \phi_2 z_t, \quad (7)$$

where z_t collects other institutional or financial variables.

Substitution yields

$$\pi_t = \phi_0 + \mu_t - g_t + \phi_1 d_t + \phi_2 z_t + u_t. \quad (8)$$

The empirical regression in equation (1) is a reduced-form approximation to this system. In particular, it does not separately identify velocity growth, fiscal deficits, external prices, or expectations.

3.2. Panel specifications

The pooled model is

$$\pi_{it} = \alpha + \beta_1 M_{it} + \beta_2 G_{it} + \beta_3 D_{it} + \varepsilon_{it}. \quad (9)$$

The individual fixed-effects specification is

$$\pi_{it} = \alpha_i + \beta_1 M_{it} + \beta_2 G_{it} + \beta_3 D_{it} + \varepsilon_{it}, \quad (10)$$

while the two-way fixed-effects model is

$$\pi_{it} = \alpha_i + \lambda_t + \beta_1 M_{it} + \beta_2 G_{it} + \beta_3 D_{it} + \varepsilon_{it}. \quad (11)$$

The supplied specification tests give the following results:

Table 2: Specification and diagnostic tests from the R workflow

Test	<i>p</i> -value	Decision at 5%	Interpretation
Breusch–Pagan LM	0.1021	Do not reject H_0	No panel effects detected
F-test FE vs OLS	0.9838	Do not reject H_0	No individual FE detected
Hausman (standard)	0.9968	Do not reject H_0	RE not rejected
Hausman (auxiliary)	0.9968	Do not reject H_0	RE not rejected
Time FE test	0.0819	Do not reject H_0	No time FE at 5%
BP heteroskedasticity	< 0.001	Reject H_0	Heteroskedasticity
White-type test	< 0.001	Reject H_0	General heteroskedasticity/nonlinearity
pbgtest	0.0075	Reject H_0	Serial correlation
Pesaran CD	0.0333	Reject H_0	Cross-sectional dependence

The formal model-selection results support pooled OLS.

This is important because the project documentation initially suggested maintaining fixed effects on theoretical grounds, but the actual production code and the exported dashboard select pooled OLS when the formal tests do not reject the pooled specification.

For methodological consistency, the report follows the estimator actually selected by the supplied code and dashboard: pooled OLS, with fixed-effects models treated as robustness specifications.

3.3. Interpretation of coefficients

The estimated pooled relationship is

$$\hat{\pi}_{it} = \hat{\alpha} + 0.436M_{it} - 0.148G_{it} + 0.377D_{it}. \quad (12)$$

The coefficient on broad money growth is measured in percentage points of inflation per percentage point of money growth.

It is not an elasticity.

For a hypothetical increase of 5 percentage points in broad-money growth, holding GDP growth and depreciation constant,

$$\Delta\hat{\pi} = 0.436 \times 5 = 2.18 \text{ pp}. \quad (13)$$

Thus the model predicts approximately 2.18 percentage points more inflation under that *ceteris paribus* change.

This is a partial-equilibrium numerical illustration, not a policy multiplier or a causal forecast.

Similarly, a 10-percentage-point depreciation is associated with

$$\Delta\hat{\pi} = 0.377 \times 10 = 3.77 \text{ pp}. \quad (14)$$

The exchange-rate channel is therefore economically large in the pooled sample.

For GDP growth, a 5 percentage point increase gives

$$\Delta\hat{\pi} = -0.148 \times 5 \approx -0.74 \text{ pp}, \quad (15)$$

but because the GDP coefficient is statistically insignificant, this arithmetic should not be interpreted as evidence of a stable GDP-inflation effect.

3.4. Sensitivity to Türkiye

The most important quantitative robustness exercise removes Türkiye.

The broad-money coefficient falls from 0.436 to 0.055:

$$\frac{0.0545}{0.436} - 1 \approx -87.5\%. \quad (16)$$

The associated 95% confidence interval is $[-0.135, 0.244]$, so zero is comfortably inside the interval.

GDP growth changes to 0.107 and remains insignificant, while depreciation remains positive at 0.275 with a confidence interval of approximately $[0.160, 0.391]$.

This is not a minor numerical adjustment.

It means that the pooled money coefficient is strongly influenced by the country with the most extreme inflation, money-growth, and depreciation observations.

The correct economic reading is therefore that there is strong regional pooled co-movement, but the evidence for a common, country-invariant money-growth effect is much weaker.

3.5. Goodness of fit

The dashboard reports $R^2 = 0.8003$ for pooled OLS, $R^2_{within} = 0.6795$ for individual fixed effects, $R^2_{within} = 0.6673$ for two-way FE, and $R^2 = 0.8003$ for random effects.

These statistics should not be compared as if they were the same object.

The pooled/RE values refer to overall variation, whereas FE values refer to within-country variation.

The high pooled R^2 is partly consistent with the strong cross-country differences in average inflation.

It does not prove that the model explains within-country inflation dynamics, nor does it prove causality.

4. Data & Empirical Context

4.1. Dataset and sample construction

The raw dataset contains 7 countries and 15 years, from 2010 through 2024, giving 105 country-year observations.

The depreciation rate is constructed as

$$D_{it} = 100 \left(\frac{ER_{it}}{ER_{i,t-1}} - 1 \right), \quad (17)$$

where ER is local currency per US dollar.

A positive value is depreciation; a negative value is appreciation.

Because the 2009 exchange-rate observations are not present, 2010 has no depreciation observation and is removed, leaving 98 observations for 2011–2024.

The regression panel is balanced:

$$7 \times 14 = 98.$$

This is a useful feature for panel diagnostics and visualization.

4.2. Descriptive evidence

Table 3: Country-level descriptive statistics, 2011–2024

Country	Mean inflation	Mean money growth	Mean GDP growth	Mean depreciation
Algeria	5.35	8.82	2.37	4.50
Egypt	14.30	19.64	3.75	18.45
Jordan	2.35	3.87	2.31	0.00
Morocco	1.89	5.53	2.97	1.37
Saudi Arabia	2.22	7.39	3.87	0.00
Tunisia	5.81	8.77	1.18	5.87
Türkiye	21.88	29.13	5.57	26.00

Several patterns stand out.

First, Türkiye is a clear high-inflation/high-money-growth/high-depreciation outlier.

Second, Egypt also combines relatively high inflation, money growth, and depreciation.

Third, Jordan and Saudi Arabia provide a useful institutional contrast because the exchange-rate variable is essentially zero under their pegs.

Fourth, Morocco has very low average inflation relative to the rest of the sample despite moderate GDP growth and relatively modest money growth.

4.3. Time dimension and major episodes

The dashboard’s time-series page shows that the strongest regional inflation acceleration occurs around 2022–2024.

The cross-country mean inflation rate rises from 4.25% in 2020 to 6.22% in 2021, 16.74% in 2022, 16.70% in 2023, and 14.61% in 2024.

Average broad-money growth also increases from 13.44% in 2020 to 16.04% in 2021 and 18.92% in 2022.

The time-series evidence therefore supports co-movement between nominal monetary expansion and inflation during the recent inflationary episode.

Nevertheless, a common shock can move both variables simultaneously.

For example, external energy and food-price shocks can raise inflation and induce policy responses that also affect money growth.

This is why the dashboard should describe the plots as evidence of association, not proof of causality.

4.4. Correlation structure

The full sample correlations calculated from the exported panel data are approximately:

Table 4: Pairwise correlations, 2011–2024

	Inflation	Money growth	GDP growth	Depreciation
Inflation	1.000	0.805	0.174	0.832
Money growth	0.805	1.000	0.204	0.681
GDP growth	0.174	0.204	1.000	0.172
Depreciation	0.832	0.681	0.172	1.000

The strong inflation–money correlation is economically suggestive but not causal.

More importantly, inflation and depreciation are even more strongly correlated (about 0.832).

This reinforces the need to control for the exchange-rate channel when evaluating the monetary variable.

When Türkiye is excluded, the inflation–money correlation falls from about 0.805 to 0.629, while inflation–depreciation remains high at about 0.821.

The monetary relationship remains visible descriptively, but it becomes materially weaker.

4.5. Stationarity and the ADF results

The country-by-country ADF tests are mostly unable to reject the null of a unit root.

Only Türkiye’s broad-money growth series is classified as stationary at the 5% level ($p = 0.0314$).

This should not be overinterpreted because each country has only $T = 14$ usable observations, giving very limited test power.

The practical implication is that stationarity should be treated as an unresolved empirical issue rather than as a solved one.

A stronger study should complement the ADF checks with panel unit-root procedures and, if persistence is substantial, use a dynamic specification such as an ARDL/PMG or dynamic panel model.

Because the dependent variable and several regressors are growth rates, the risk of classic level-variable spurious regression is lower than in a levels-on-levels specification, but serial dependence and persistent common shocks remain important.

4.6. Panel specification tests and diagnostics

The LM test ($p = 0.1021$) and the FE-vs-pooled F-test ($p = 0.9838$) do not support country-specific intercepts statistically.

The Hausman tests also do not reject random-effects consistency, although the pooled model is already selected before that stage because the panel-effect tests do not reject pooling.

The time-effects test has $p = 0.0819$, so time effects are not statistically significant at the 5% threshold.

The more serious issues are in the error structure.

Heteroskedasticity is detected by both the Breusch–Pagan and White-type tests.

Serial correlation is detected by the panel Breusch–Godfrey test, and Pesaran’s CD rejects the null of no cross-sectional dependence.

These results justify the use of robust inference.

The supplied workflow responds with both country-clustered HC3 standard errors and Driscoll–Kraay standard errors.

4.7. Dashboard interpretation, page by page

Dashboard Page 1 – cross-country level.

The first page correctly communicates the cross-sectional heterogeneity in mean inflation.

Türkiye is clearly the highest-inflation country, followed by Egypt, while Morocco, Jordan, and Saudi Arabia are much lower.

The table at the bottom is especially useful because it places inflation, GDP growth, broad-money growth, and depreciation side-by-side.

A methodological improvement would be to label all reported figures explicitly as “2011–2024 means” because the overall project title still says 2010–2024 while the analysis page actually excludes 2010.

Dashboard Page 2 – time-series dynamics.

The four trend charts are useful for identifying turning points.

The strongest common message is the sharp rise in inflation and money growth around 2021–2024, with exceptional movements in Türkiye and Egypt.

The depreciation chart is especially informative because it visually separates pegged regimes from flexible or managed exchange rates.

However, the many overlapping lines make country-level interpretation difficult without filtering.

A cleaner final dashboard would provide a small-multiple or selectable-country view as the default.

Dashboard Page 3 – bivariate relationships.

The positive slope between broad-money growth and inflation is an appropriate descriptive result, but the sentence “consistent with the Quantity Theory of Money” should be followed by an explicit warning that a bivariate slope does not identify causality.

The same page strongly suggests that outliers in Türkiye drive part of the slope.

The scatterplots are therefore most useful as motivation for the sensitivity analysis.

Dashboard Page 4 – model results.

The coefficient panel is the analytical core.

Broad money is positive and significant, depreciation is positive and significant, and GDP growth is insignificant.

The page also reports the country fixed effects and several R^2 measures.

These fixed effects should be labeled “supplementary FE estimates” rather than presented as if the FE model were the selected estimator.

The bottom-right text correctly identifies pooled OLS as selected by the formal tests.

Dashboard Page 5 – specification and diagnostics.

This page is valuable because it demonstrates that the inference problem is not trivial.

Heteroskedasticity, serial correlation, and cross-sectional dependence are all detected.

The residual-versus-fitted panel also shows that the largest residuals tend to occur in Türkiye, especially around the high-inflation episode.

The actual-versus-fitted chart shows that pooled OLS captures broad country differences but has difficulty fitting sharp country-specific inflation spikes.

4.8. Visual and econometric adjustments recommended

The project is already structurally strong, but several changes would improve its academic quality.

1. **Align the estimator narrative.** The R script comments should not say that FE is “maintained on theoretical grounds regardless of tests” when the active code actually selects pooled OLS if both panel tests fail to reject.

Choose one rule and apply it consistently.

The current dashboard follows the formal-test rule, so the report does the same.

2. **Rename the time scope.** The raw data cover 2010–2024, but the econometric sample is 2011–2024.

Every regression page, table, and figure should say 2011–2024 unless it explicitly shows raw data.

3. **Separate descriptive and causal language.** Replace wording such as “impact” or “effect” with “association” unless a stronger identification strategy is introduced.

The title can remain the project’s research question, but the empirical section should be explicit about the identification limitation.

4. **Make the Türkiye sensitivity result central, not secondary.** The coefficient falls by roughly 87.5% when Türkiye is removed.

This is the most important robustness fact in the current analysis.

5. **Report robust inference consistently.** Because the diagnostics detect three forms of dependence, the main table should display either Driscoll–Kraay inference or a clearly justified country-clustered approach, with the other reported as robustness.

6. **Consider dynamics.** Inflation and monetary growth may respond to their own past values.

A lagged model would better match the transmission mechanism:

$$\pi_{it} = \rho\pi_{i,t-1} + \beta M_{it-1} + \dots$$

7. **Add omitted macro controls where data permit.** Fiscal balance or government deficit, policy interest rate, oil/energy prices, food-price shocks, and global inflation would materially improve the model’s economic interpretation.

5. Policy Recommendations & Conclusion

5.1. Monetary policy

The positive pooled money-growth coefficient supports close monitoring of broad money, credit expansion, and deposit growth as inflation indicators.

However, a simple mechanical rule of “reduce M2 growth because inflation rises” would be inappropriate.

The evidence indicates association, and the Türkiye sensitivity result shows that the coefficient is not stable across countries.

A better policy framework is to monitor broad money jointly with inflation expectations, credit growth, exchange-rate conditions, output gaps, and fiscal developments.

Monetary tightening should be proportionate to the source of inflation: demand-driven inflation is more directly addressed by tighter financial conditions, whereas supply-driven inflation requires complementary supply-side and fiscal measures.

5.2. Exchange-rate policy

The 0.377 depreciation coefficient and the high inflation–depreciation correlation indicate that exchange-rate stability can be an important inflation-stabilization channel.

Countries with flexible exchange rates should monitor second-round effects from depreciation into wages and administered prices.

Pegged regimes, in turn, require sufficient reserves, fiscal credibility, and monetary consistency to remain credible.

Exchange-rate defense should not substitute indefinitely for the correction of underlying macroeconomic imbalances.

5.3. Fiscal-monetary coordination

A macroeconomic framework focused only on money growth can miss fiscal drivers of inflation.

Large fiscal deficits can sustain aggregate demand, increase financing needs, and affect expectations about future monetary conditions.

Monetary credibility is therefore strengthened when fiscal policy is consistent with the inflation objective.

The relevant policy question is not only how much money grows, but why it grows and whether the expansion is compatible with expected output growth and fiscal sustainability.

5.4. Supply-side and regulatory policies

Because food, energy, imported intermediates, and exchange rates are important for MENA inflation, supply-side resilience matters.

Policies that improve competition, logistics, domestic food production, energy efficiency, and import diversification can reduce the persistence of cost shocks.

Temporary targeted support may protect vulnerable households, but generalized subsidies can weaken fiscal positions and delay price adjustment.

5.5. Policy trade-offs and unintended consequences

Tighter monetary conditions can lower inflation but may reduce investment, employment, and credit access.

Exchange-rate stabilization can support prices but may consume reserves or reduce export competitiveness.

Fiscal consolidation can improve credibility but weaken aggregate demand in the short run.

Broad subsidies can lower measured inflation temporarily but may increase deficits and create distortions.

The optimal response is therefore country-specific and should balance price stability against output and financial-stability risks.

5.6. Conclusion

This empirical project provides meaningful evidence of a positive regional relationship between broad-money growth and inflation.

In the pooled model, a one-percentage-point increase in broad-money growth is associated with about 0.436 percentage points of higher inflation, conditional on GDP growth and currency depreciation.

The depreciation coefficient is also large and significant, while GDP growth is not statistically significant.

Yet the most important conclusion is the robustness caveat.

Türkiye accounts for a disproportionate share of the high-inflation/high-money-growth observations.

Excluding it reduces the money coefficient to about 0.055 and makes it statistically insignificant.

The evidence therefore does **not** justify the statement that broad money has a uniform causal impact of 0.436 on inflation in all seven MENA countries.

Economically, the dashboard is strongest when interpreted as a comparative macroeconomic diagnosis: monetary expansion is positively associated with inflation, but its effect depends on exchange-rate regimes, institutional credibility, real-side conditions, and country-specific shocks.

The next empirical step should be a dynamic panel framework with stronger treatment of common shocks, more macroeconomic controls, and a strategy that distinguishes monetary endogeneity from genuine predictive or causal effects.

Technical Appendix: Reproducibility and Reporting Notes

Variables

- Inflation: CPI inflation rate, percent.
- Broad money growth: annual growth rate of the project’s broad-money measure, percent.
- GDP growth: annual real GDP growth, percent.
- Exchange rate: local currency units per US dollar.
- Depreciation rate:

$$100 \left(\frac{ER_t}{ER_{t-1}} - 1 \right).$$

Sample

7 countries $\times$ 15 raw years = 105 observations.

After losing 2010 because of the lagged exchange-rate transformation:

$$7 \times 14 = 98$$

regression observations.

Robustness evidence

Table 5: Full sample versus Türkiye-excluded sensitivity

Variable	Full sample	Without Türkiye	Change / interpretation
Broad money growth	0.436	0.055	Strong attenuation; CI includes zero
GDP growth	-0.148	0.107	Remains insignificant
Depreciation	0.377	0.275	Positive effect remains robust

Important interpretation rule

The supplied dashboard and exports use the selected estimator “Pooled OLS”.

Fixed-effects and two-way fixed-effects models are reported as supplementary specifications.

Any future version of the report should preserve this consistency unless the model-selection rule in the R script is intentionally changed.

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